

Research Notes on Denoising diffusion probabilistic model

Denoising diffusion probabilistic model

>> Section 1

$d \phi_t(x) = v_t(\phi_t(x)) dt \in [0, 1]$, starting from $\phi_0(x) = x$ $\frac{d}{dt} \phi_t(x) = v_t(\phi_t(x))$ $\phi_t(x) \in [0, 1]$, $\phi_0(x) = x$

>> Section 2

The data pair (x_0, x_1) can be any coupling of π_0 and π_1 , typically independent (i.e., $(x_0, x_1) \sim \pi_0 \times \pi_1$) obtained by randomly combining observations from π_0 and π_1 . This process ensures that the trajectories closely mirror the density map of x_t trajectories but reroute at intersections to ensure causality.

>> Section 3

$\nabla_x \ln q(x_t) = \frac{1}{\sigma_t^2} (-x_t + \alpha^{-1} E q[x_0 | x_t])$ $\nabla_x \ln q(x_t) = \frac{1}{\sigma_t^2} (-x_t + \alpha^{-1} E q[x_0 | x_t])$

>> Section 4

Architectural improvements proposed various architectural improvements. For example, they proposed log-space interpolation during backward sampling. Instead of sampling from $x_{t-1} \sim N(\mu_t(x_t, x_0), \sigma_t^2)$, they recommended sampling from $N(\mu_t(x_t, x_0), (\sigma_t^2 v_t^2))$ for a learned parameter v_t . In the v -prediction formalism, the noising formula $x_t = \alpha^{-1} x_0 + 1 - \alpha^{-1} \epsilon$ is reparameterised by an angle ϕ_t such that $\cos \phi_t = \alpha^{-1}$ and a "velocity" defined by $\sin \phi_t = \epsilon$. The network is trained to predict the velocity $\hat{v}_t$, and denoising is by $x_{\phi_t - \delta} = \cos(\delta) x_{\phi_t} - \sin(\delta) \hat{v}_t$. This parameterization was found to improve performance, as the model can be trained to reach total noise (i.e. $\phi_t = 90^\circ$) and then reverse it, whereas the standard parameterization never reaches total noise since $\alpha^{-1} > 0$ is always true.

>> Section 5

$N(\mu, \Sigma)$ is the normal distribution with mean μ and variance Σ , and $N(x | \mu, \Sigma)$ is the probability density at x . A vertical bar denotes conditioning. A forward diffusion process starts at some starting point $x_0 \sim q$, where q is the probability distribution to be learned, then repeatedly adds noise to it by $x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} z_t$ where $z_1, \dots, z_T$ are IID (Independent and identically distributed random variables) samples from $N(0, I)$. The coefficients $\sqrt{1 - \beta_t}$ and $\sqrt{\beta_t}$ ensure that $\text{Var}(X_t) = I$ assuming that $\text{Var}(X_0) = I$. The values of β_t are chosen such that for any starting distribution of x_0 , if it has finite second moment, then $\lim_{t \rightarrow \infty} x_t | x_0$ converges to $N(0, I)$. The entire diffusion process then satisfies $q(x_{0:T}) = q(x_0) q(x_1 | x_0) \dots q(x_T | x_{T-1}) = q(x_0) N(x_1 | \alpha_1 x_0, \beta_1) \dots N(x_T | \alpha_T x_{T-1}, \beta_T)$ or $\ln q(x_{0:T}) = \ln q(x_0) - \sum_{t=1}^T \frac{1}{2\beta_t} \|x_t - \sqrt{1 - \beta_t} x_{t-1}\|^2 + C$ where C is a normalization constant and often omitted. In particular, we note that $x_{1:T} | x_0$ is a Gaussian process, which affords us considerable freedom in reparameterization. For example, by standard manipulation with Gaussian process, $x_t | x_0 \sim N(\alpha^{-1} x_0, \sigma_t^2)$

>> Section 6

Simple, $t = E x_0, x_t \sim q[\theta(x_t, t) - x_t - \alpha^{-1} x_0 \sigma_t^2] = E x_t \sim q(\cdot | x_t) [\theta(x_t, t) - x_t - \alpha^{-1} x_0 \sigma_t^2]$ $L_{\text{simple}, t} = E_{x_0, x_t \sim q} [\frac{\|x_t - \sqrt{1 - \beta_t} x_{t-1}\|^2}{\sigma_t^2} | \theta(x_t, t) - x_t - \alpha^{-1} x_0 \sigma_t^2]$